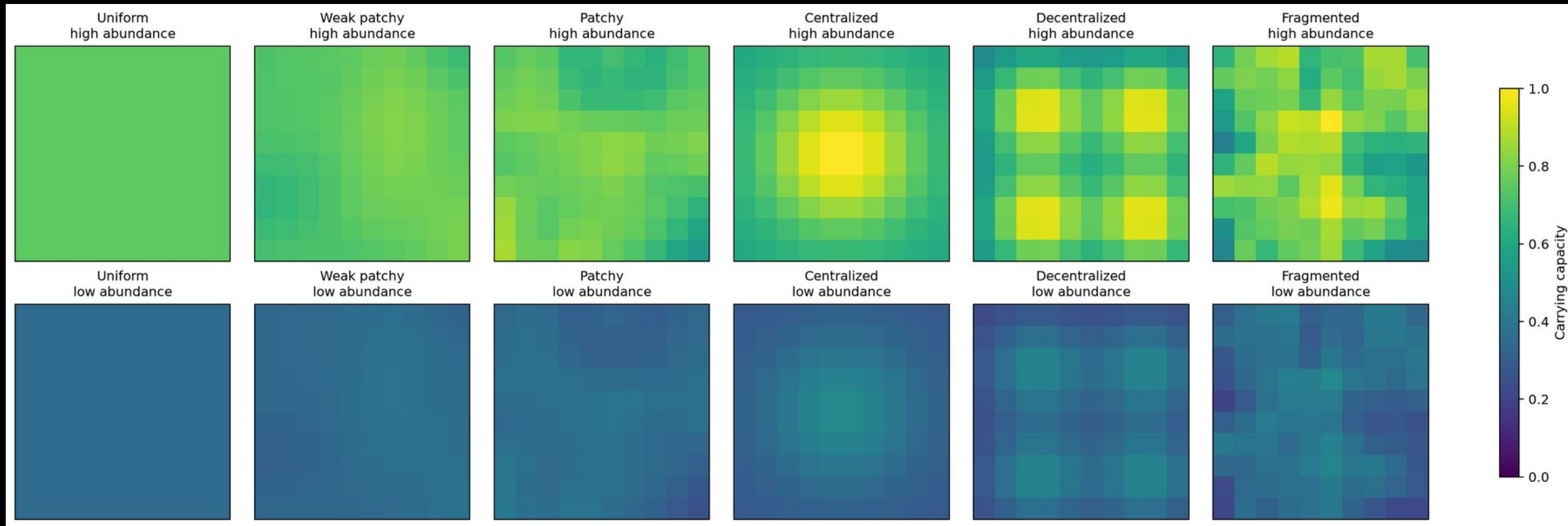
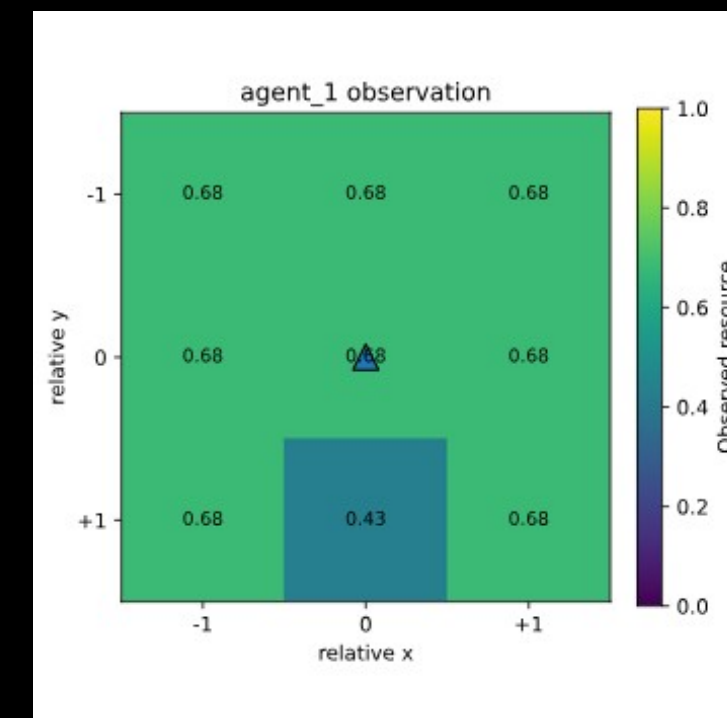




Technology, Access, and Welfare in Artificial Societies

Social-Technical Multi-Agent Environments



Roadmap

1.baseline resource world, (September)

1.Inheritance, tool-building, trade, and social learning.(October)

1.Build and run scenarios (November)

1.Write, analyze and share results (December)

Inequality, Welfare & Cooperation

Social Learning

Research question

Can agents that locally predict the behavior of their information sources use prediction error to adapt whom they observe, and can that decentralized rewiring improve or destabilize the sustainability of a common-pool resource?

How does the effect of rewiring depend on ecological conditions?

Can ecological shocks, or centralized, discrete changes trigger rewiring and collective changes in local prediction and behavior?

Initialization

Results

Recap

Baseline resource world (September)

- Built tile-world with renewable resources and ecological mechanisms for resource regeneration
- Added agents with random actions & Q-learning*
- Meeting with Anil Yaman for feedback, implementation questions and directional clarity
- Currently revising code based on meeting

*Still being implemented after meeting with
Anil

Broadly

How does environment effect collective welfare and outcomes?

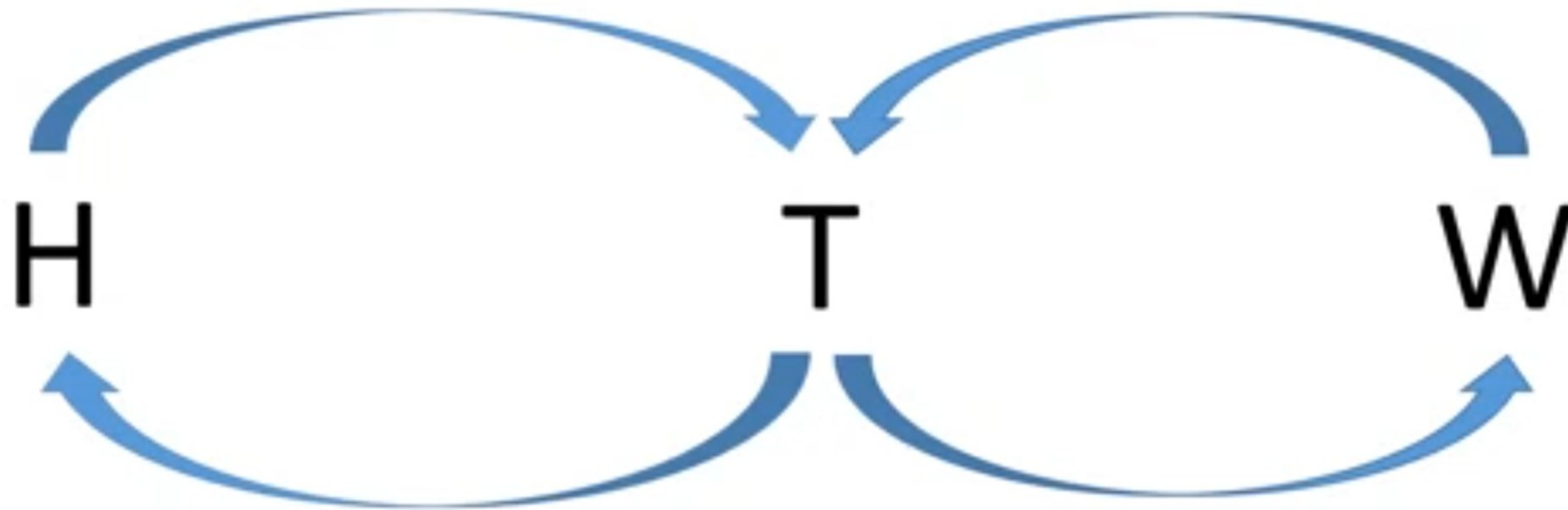
How does environment mediate individual decision-making?

How does environment affect individual preference to cooperate for collective goals?

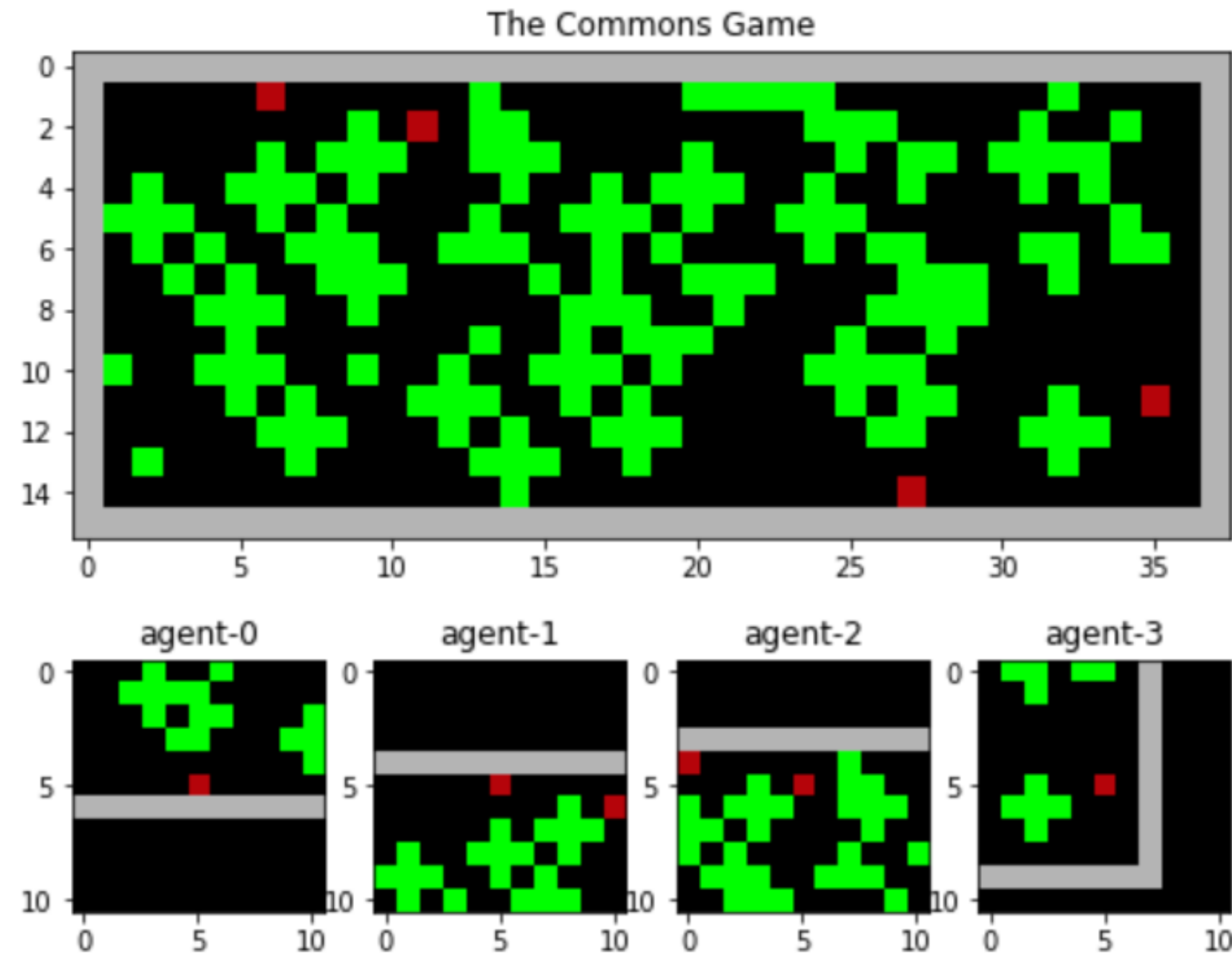
Environment Shapes World Perceptions

Technological mediation of morality

1. Helping to form (moral) perceptions about the world
2. Helping to shape situations of (moral) choice
3. Co-shaping our values and even changing them



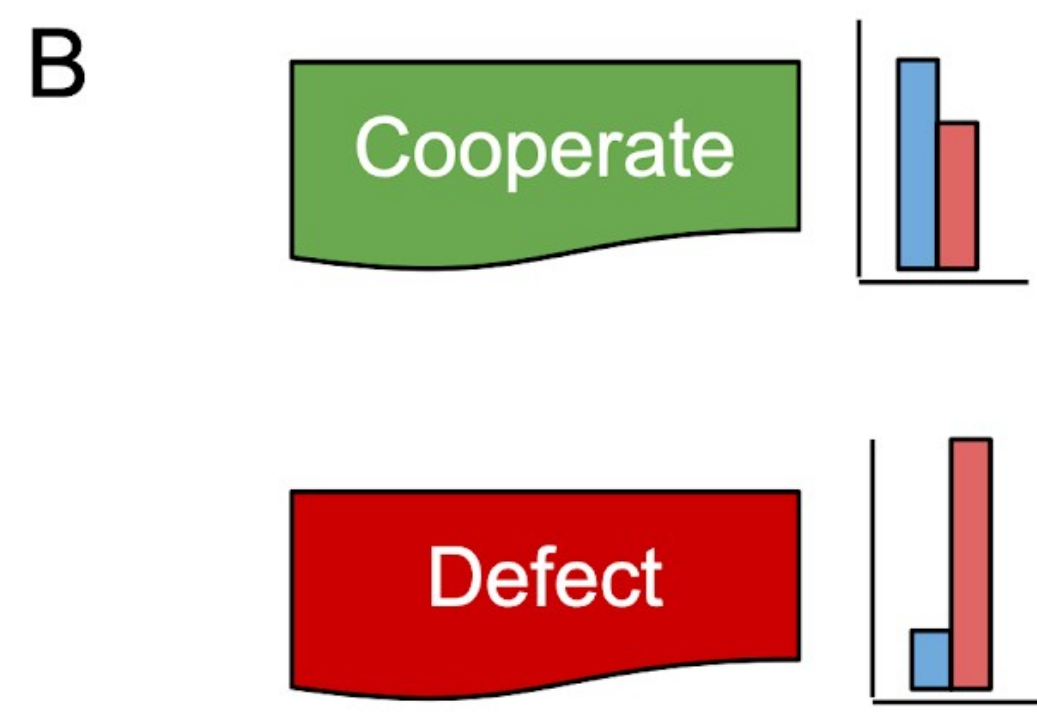
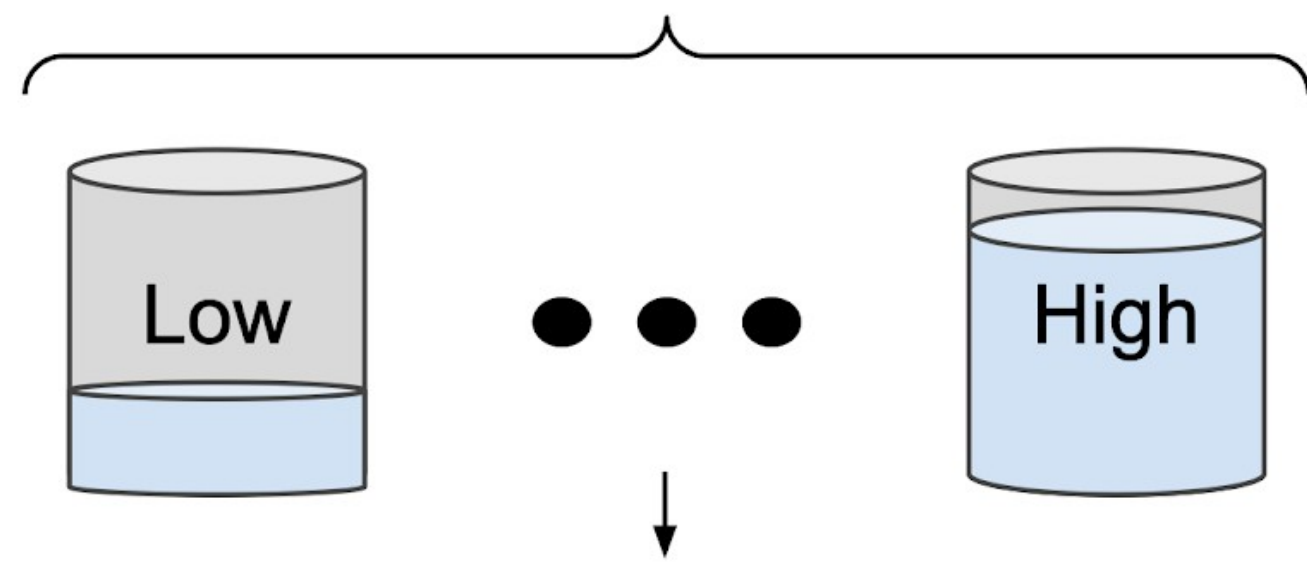
Commons Game



- Players collect shared resources (“apples”).
- Apple regrowth depends on nearby unharvested apples.
- If a local area is fully depleted, apples cannot regrow until reset.
- Individual incentives favor rapid harvesting, while collective welfare requires restraint.
- Simultaneous harvesting increases the risk of irreversible local depletion.

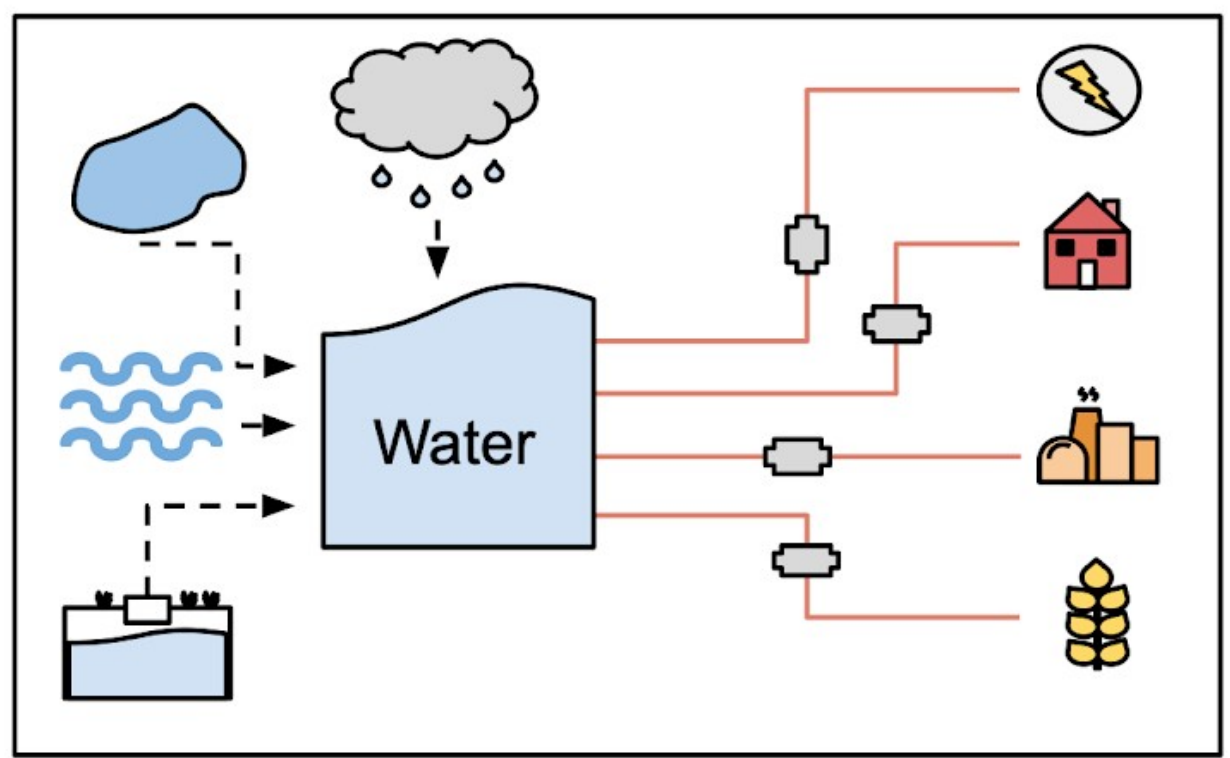
Common-Pool Resource Games

A Regeneration Rates



C

Evaluate



First Scenario

What welfare dynamics in agent populations arise from renewable environments alone?

Representing Ecologically Dynamic Environments

The tile world uses a surplus-production model to represent the self-regulating dynamics of an ecosystem

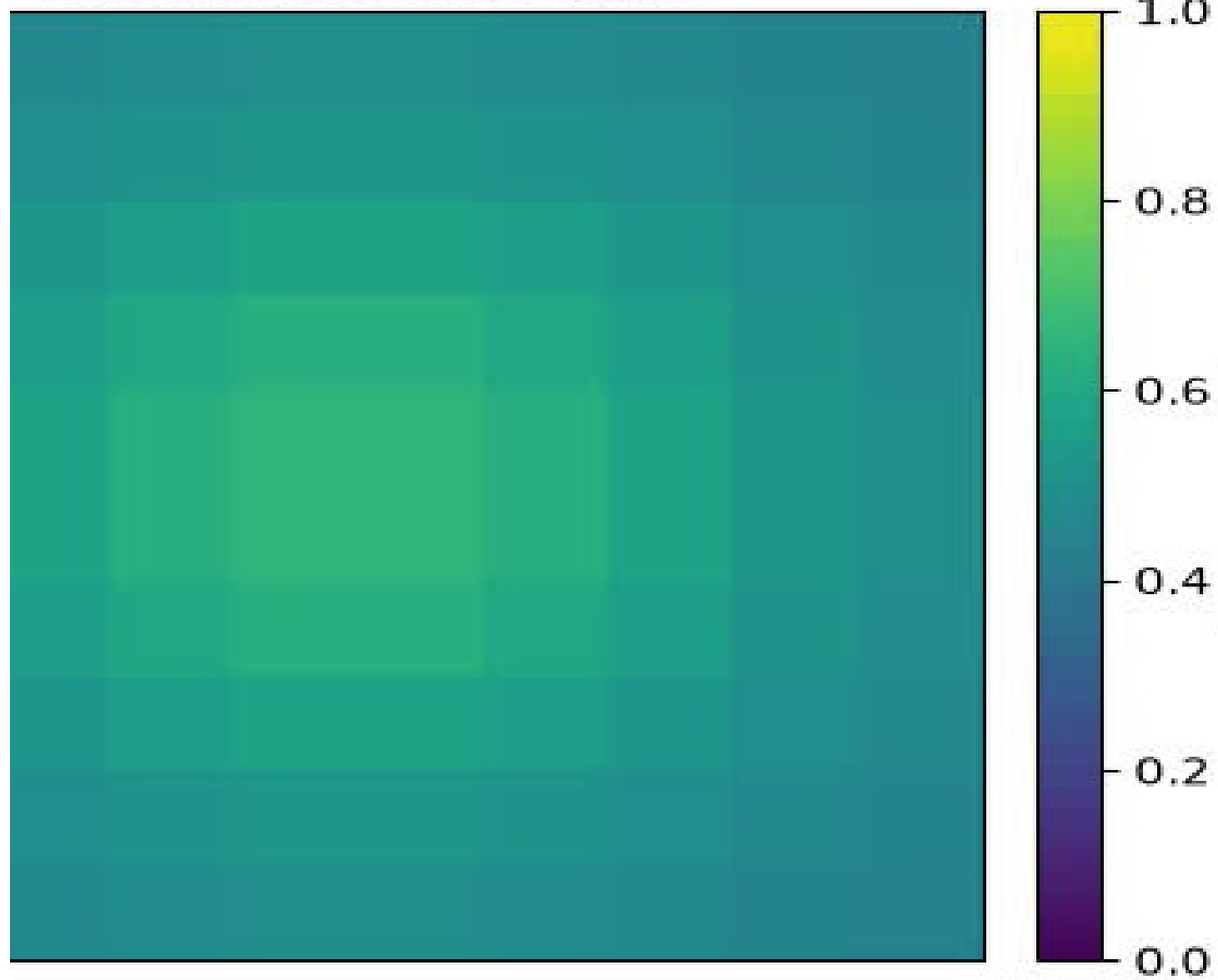
Changes in one tile will cause redistribution the other tile's states

Tile Properties: Carrying capacity, resource regeneration, proportional depletion, and nearest-neighbor diffusive coupling.

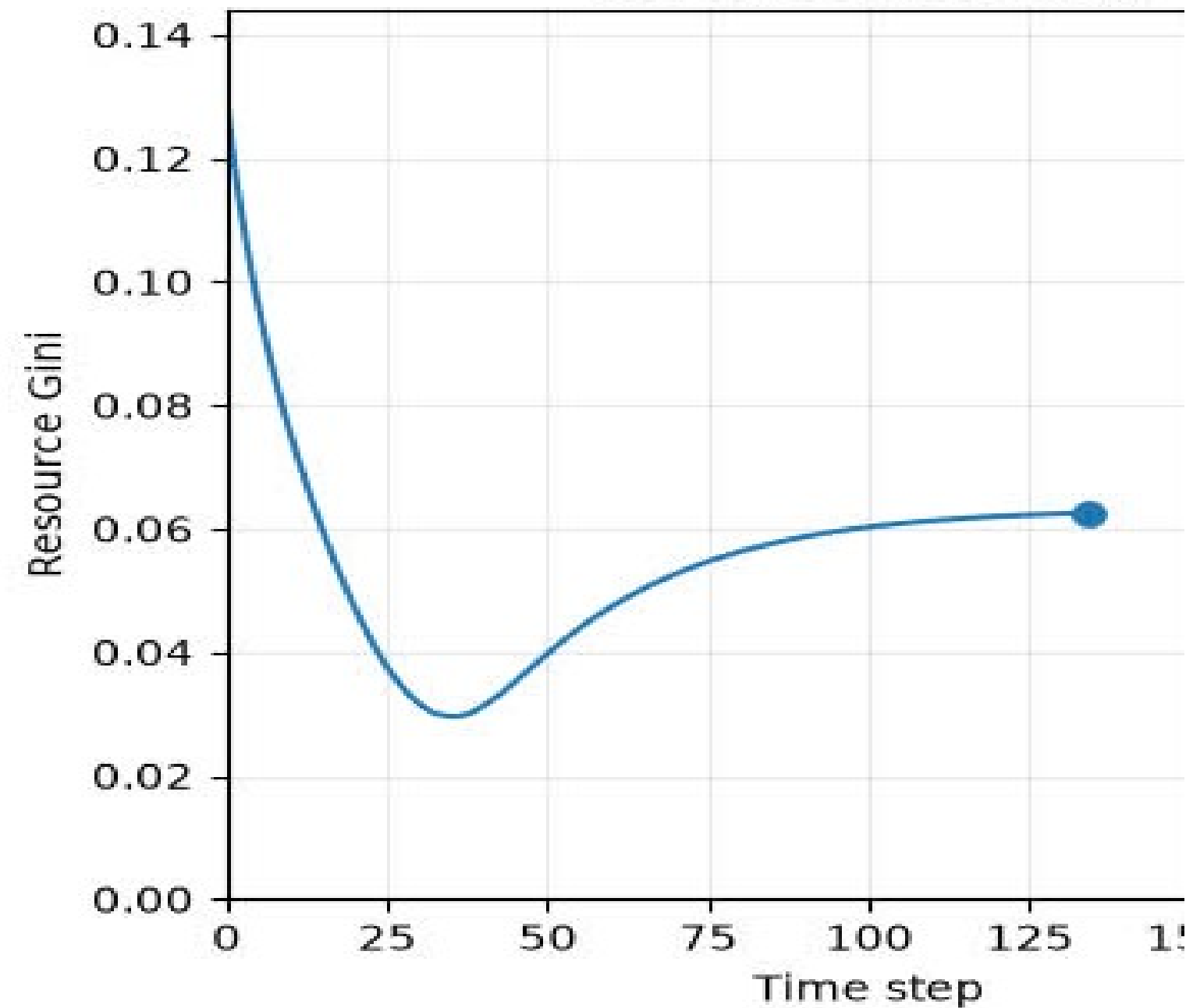
Creating a Environmental Equilibrium

centralized_high | shock | step 134
Resource inequality: 0.063

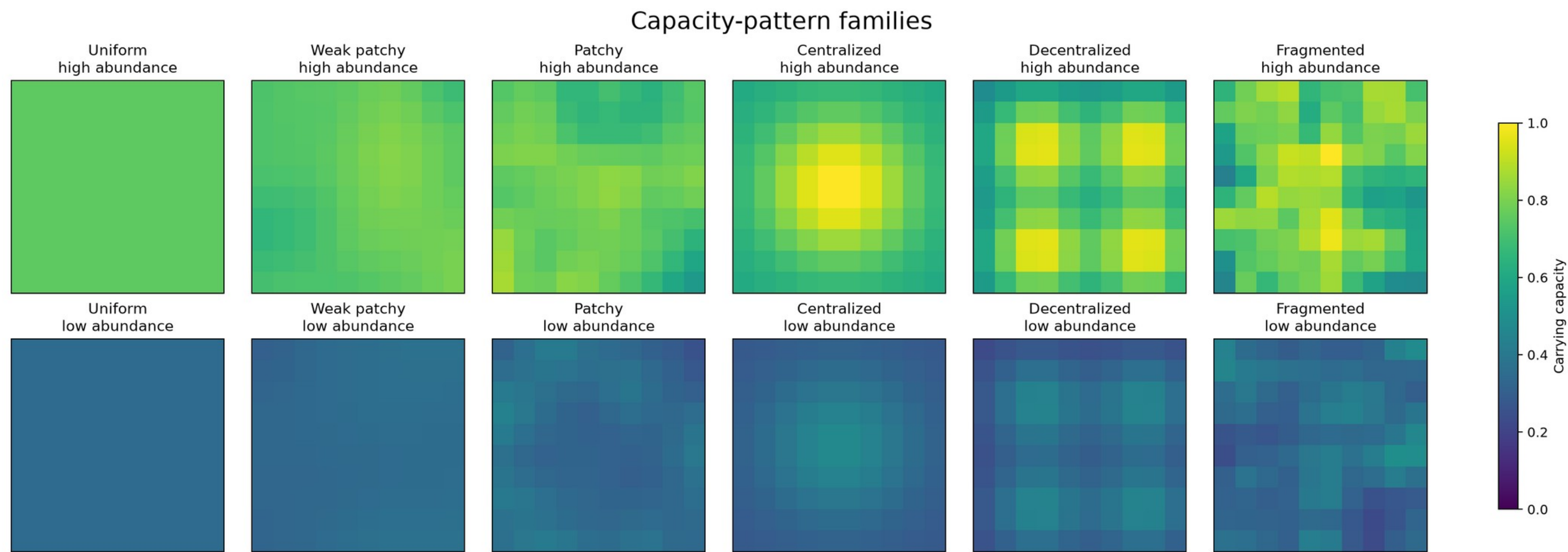
Current resource field



Resource inequality



Multiple Environments with Differing Equilibriums

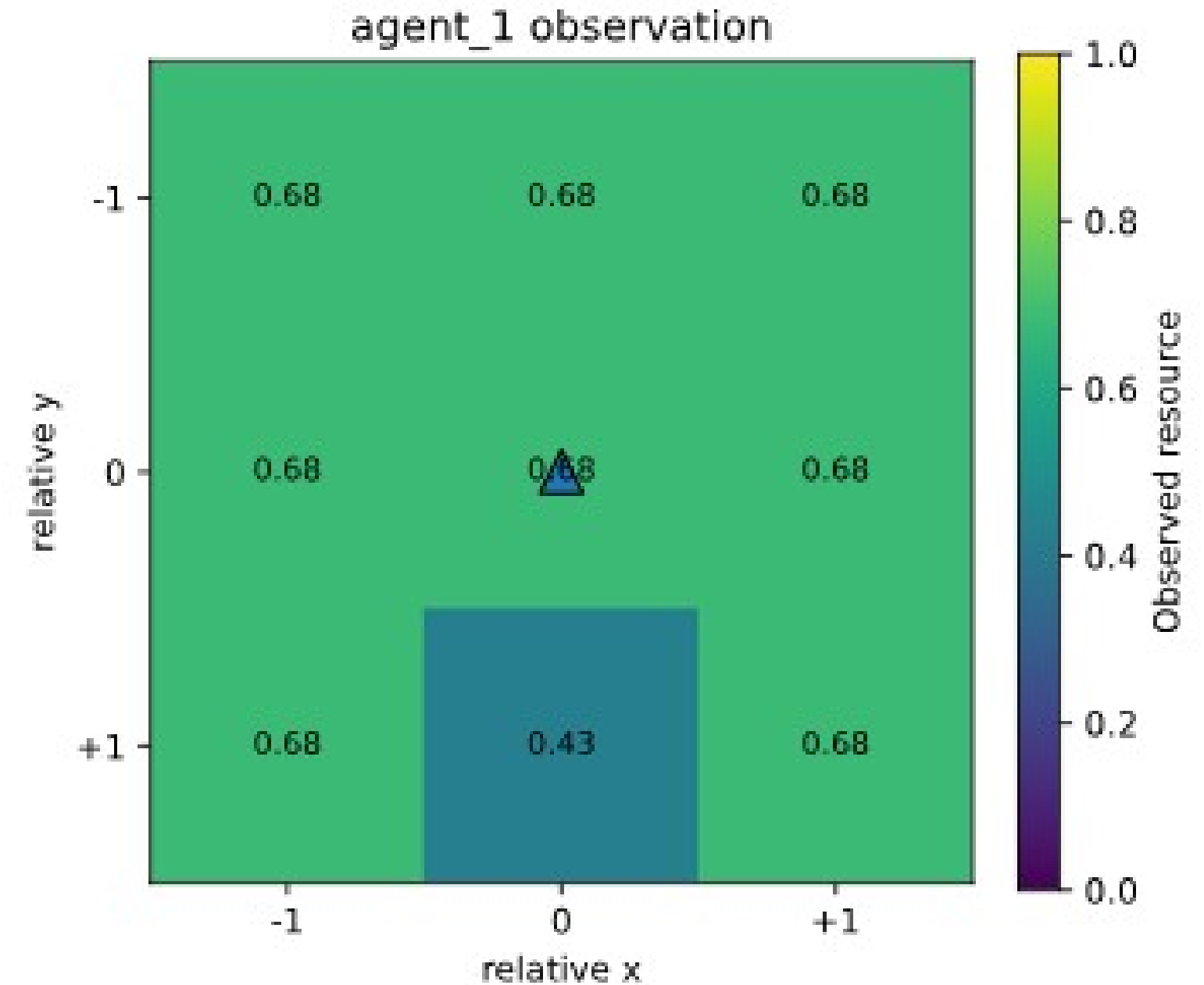


Capacity is the amount of resource available on each tile

Added random agent actions (No feedback from behavior)

```
# Actions  
STAY = 0  
UP = 1  
DOWN = 2  
LEFT = 3  
RIGHT = 4  
HARVEST = 5
```

```
moves = {  
    UP: (0, -1),  
    DOWN: (0, 1),  
    LEFT: (-1, 0),  
    RIGHT: (1, 0),  
}
```



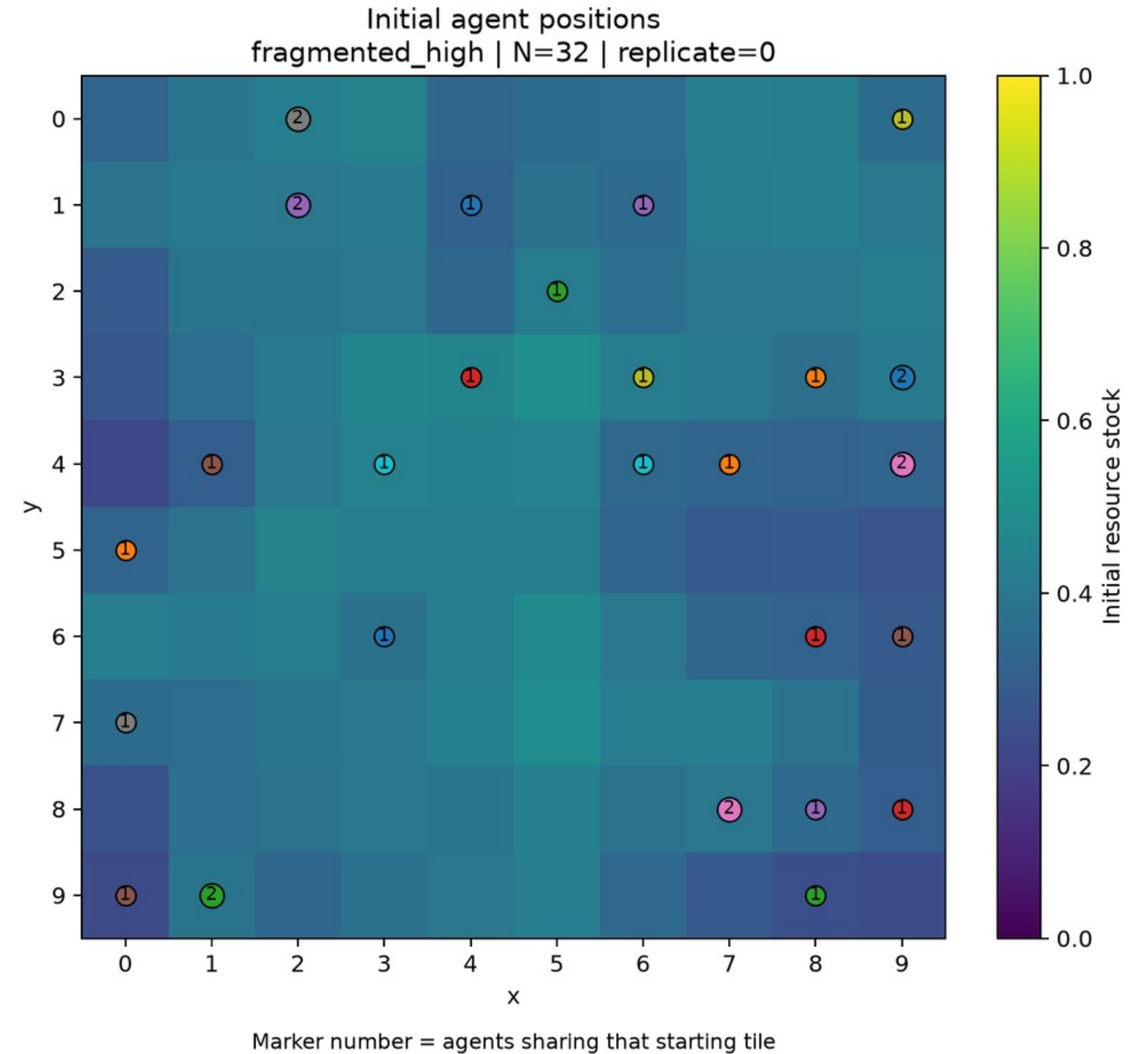
Measuring inequality and welfare

Gini Coefficient is used to measure how unequally resource stock is spread across tiles

$$G = \frac{\sum_i \sum_j |R_i - R_j|}{2n \sum_i R_i}$$

All agent initialization were random
1000 Steps per
Population
Sizes:

$$N \in \{4, 8, 16, 32\}$$

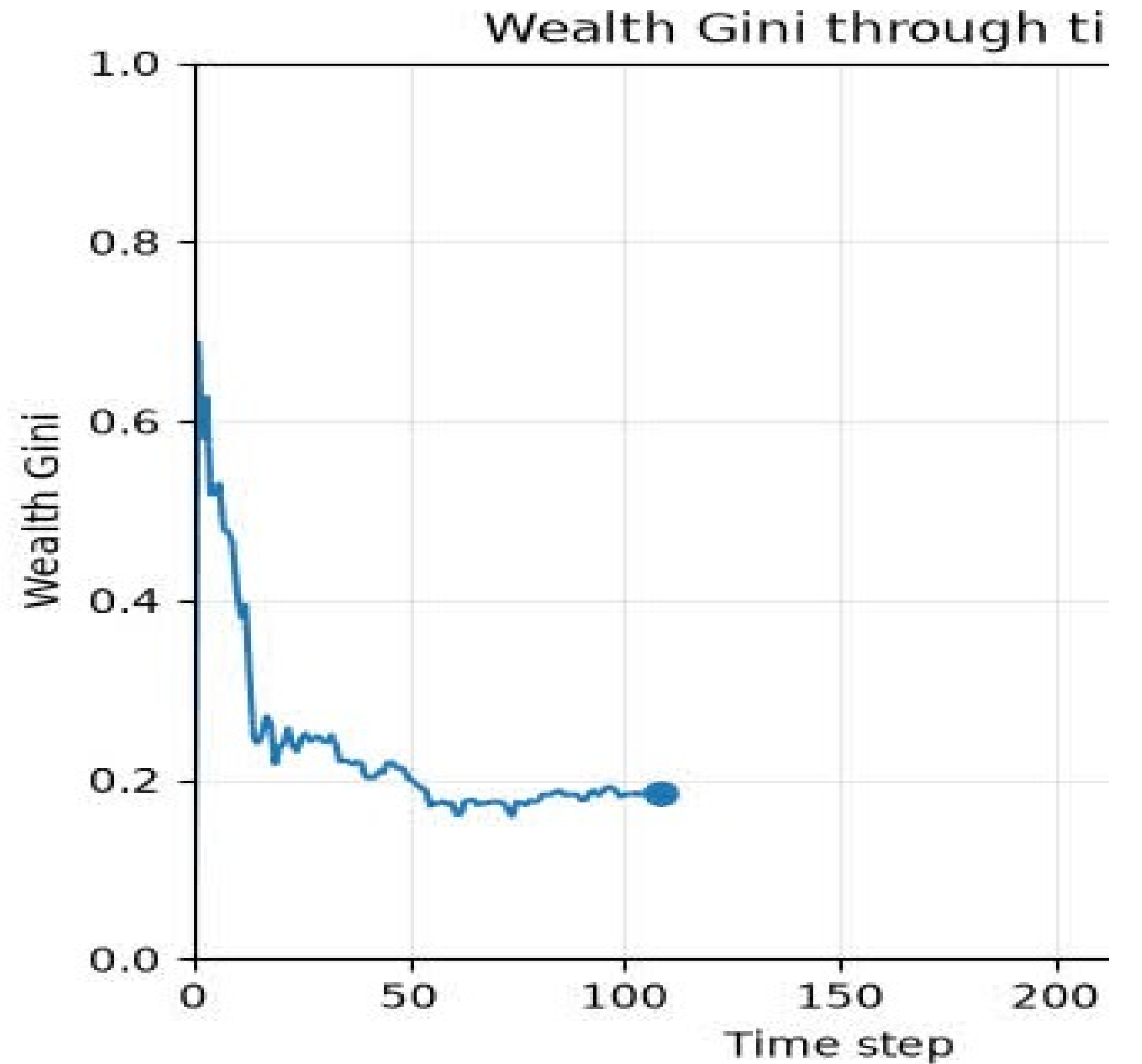
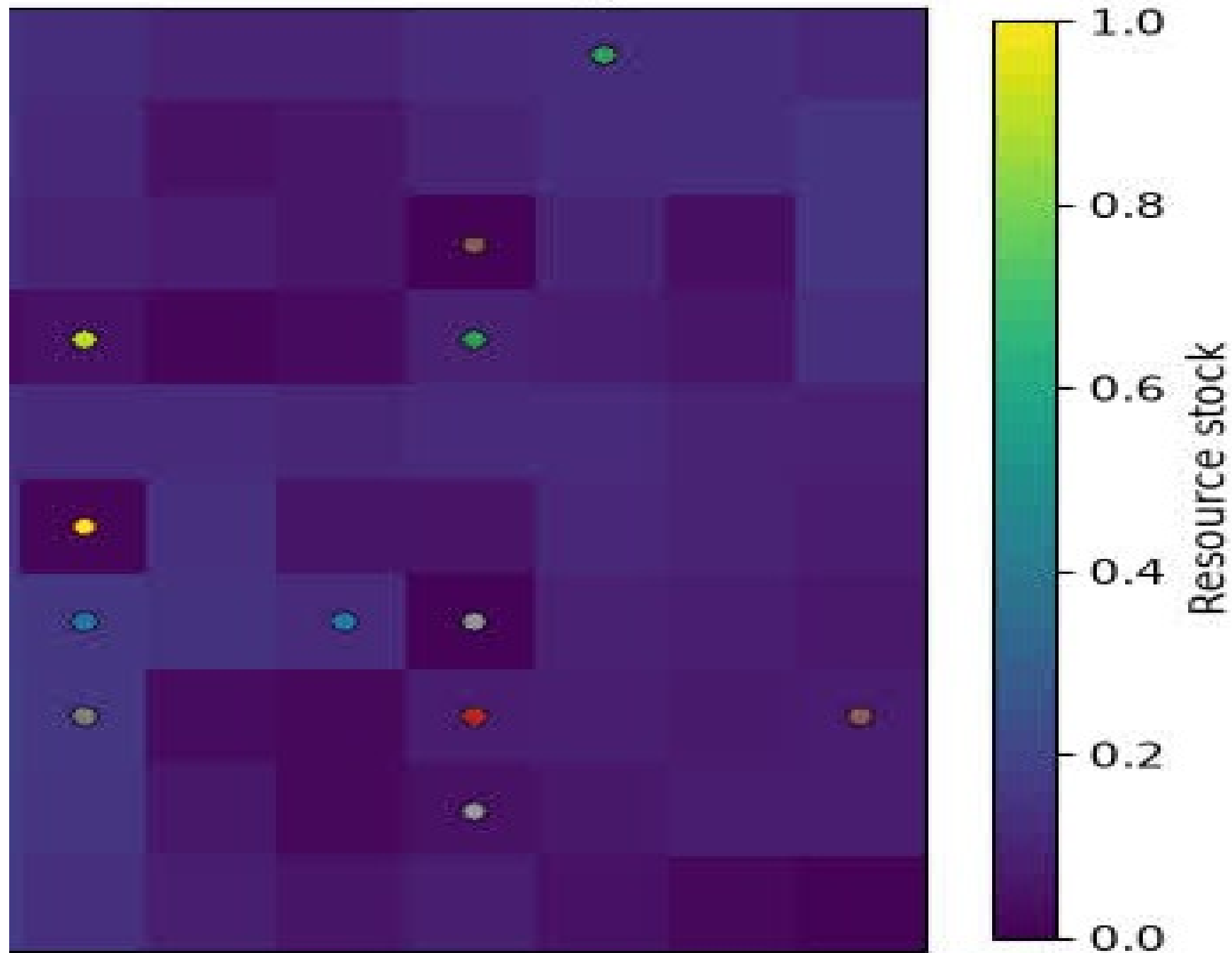


A higher gini coefficient means more inequality among individuals

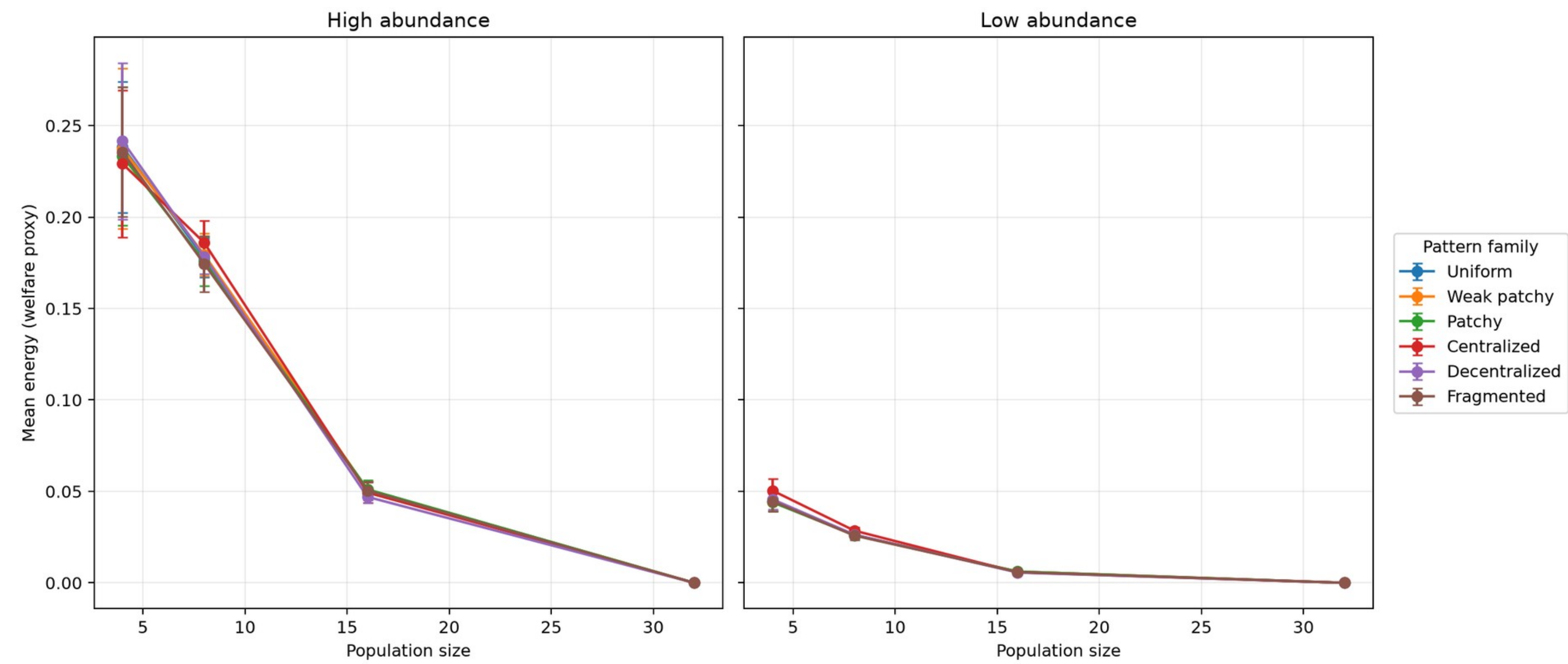
Agents alter Environment Ecology

patchy_low | N=16 | step=108

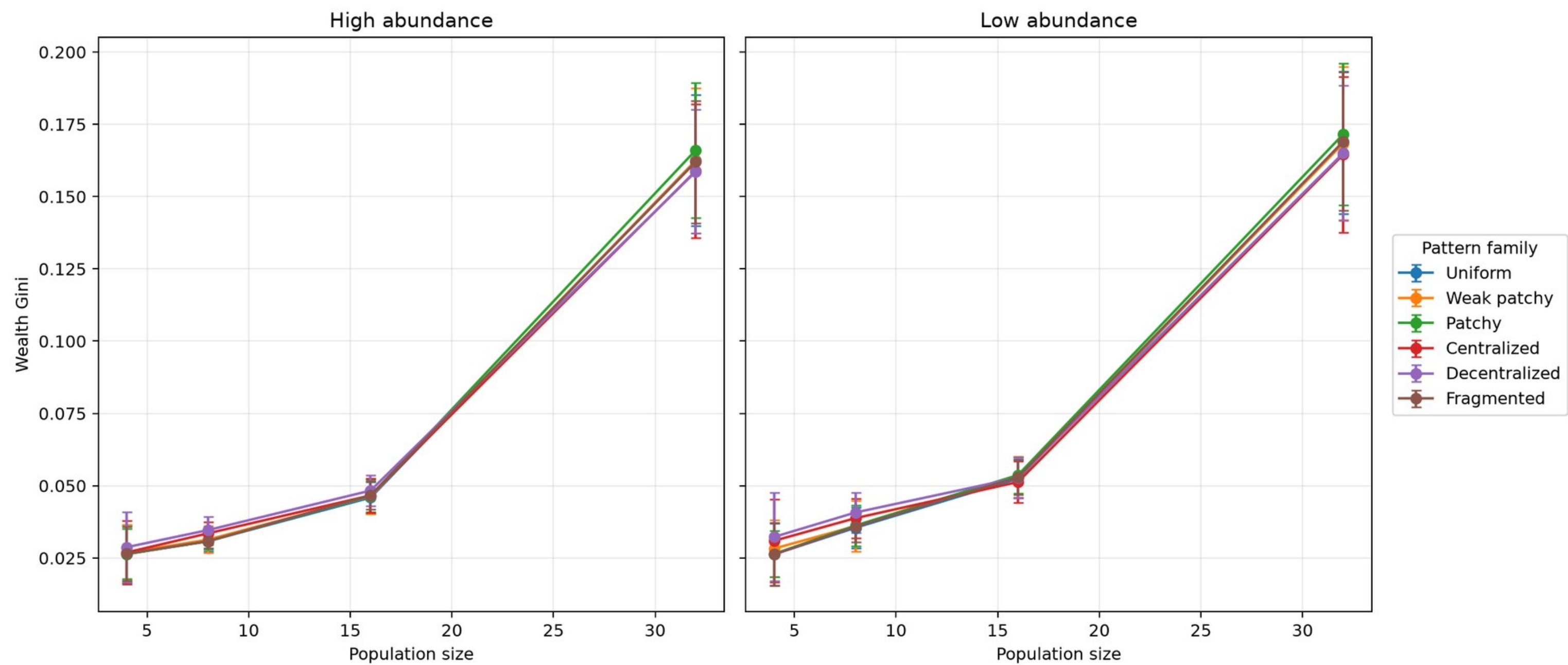
ce field + random agents



Population pressure led to loss in welfare in random agents



Similar Inequality Trends for High & Low Abundance



Met with Anil Yaman

- Constraint model to fundamental mechanisms
- Adding spatial movement for agents is not necessary for model
 - Focus only on how agents extract resource (Cooperate or Defect)
- Formulate baselines to test cooperation in different ecological environments where agents learn to adapt to their ecological setting and how that effects cooperation collectively
- How to extend project for cultural intersections
- Compare runs in singular and mixed environments(High & low abundance together with different spatial organizations)

Baseline Questions

1.What extraction policies do agents learn from environment alone?

1. Do local conditions predict collective behavior?

1.Can agent's learn collective behavior's to improve welfare?

1.How does environmental heterogeneity change policies and outcomes?

Experiment

Two Types of Agent Action

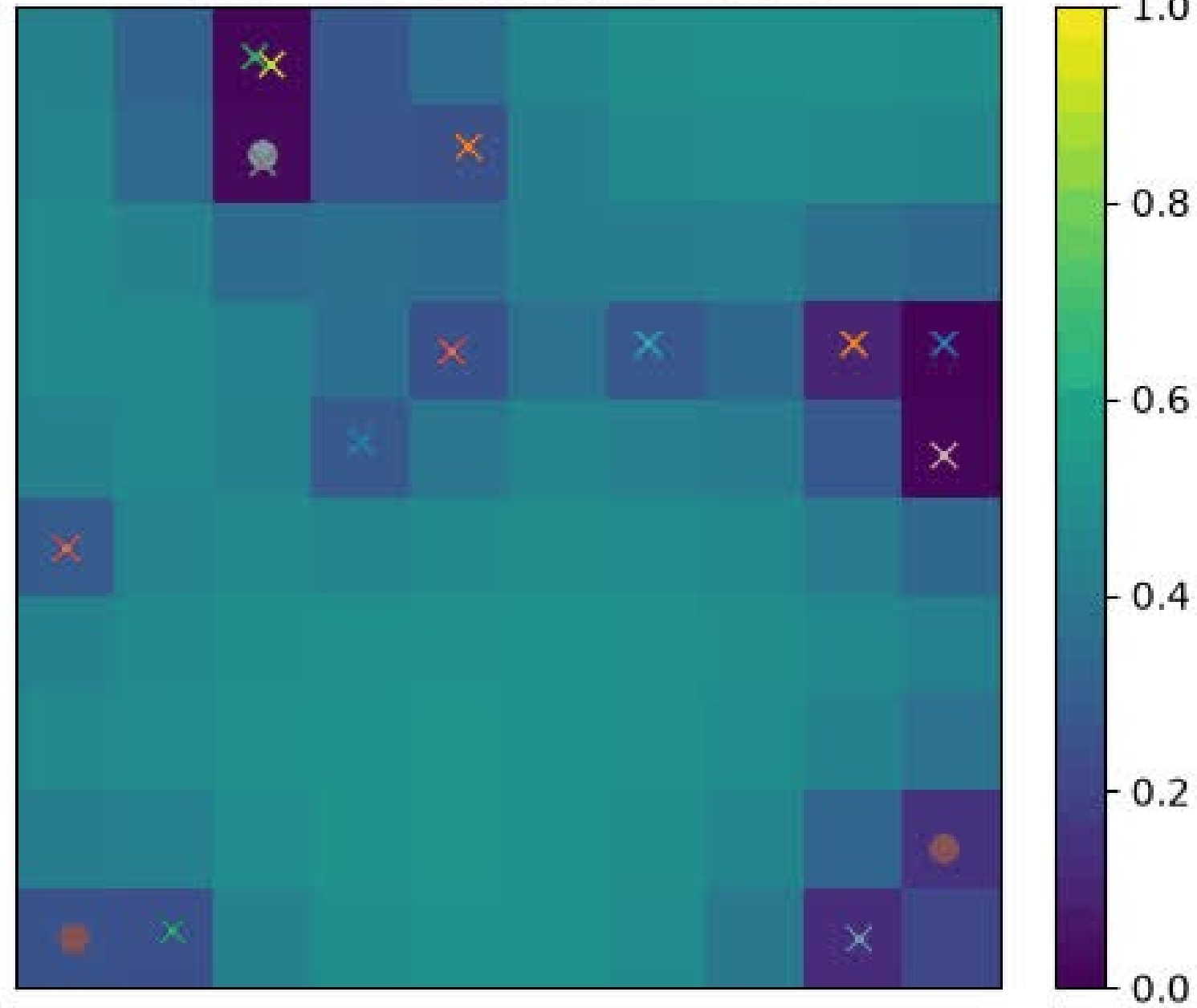
- Low Extraction (Cooperation)
- High Extraction (Defection)

1. Train independent
2. Freeze and evaluate in the ecology observed during training.
3. Reset parameters and evaluate the same learned policy again from a initial condition.
4. From the same initial condition evaluate three controls:
 - always low extraction (Agents always cooperate);
 - always high extraction (Agents always defect);
 - random 50/50 extraction (Equal chance of cooperating or defecting).
5. Record tile state occupancy and realized state

Simplified Model

uniform_high | N=16 | step=225

Resource field and actions
circle = cooperate, x = defect



Collective learning dynamics

